

Homework 1

EN.601.639 | Fall 2026

You must submit either a physical handwritten homework or print out your homework to submit at the beginning of class.

Problem 1: PCA for Different Distributions

- a) Write out the PDF of a negative binomial distribution? When would you use a negative binomial distribution versus a Poisson distribution?
- b) Write out the GLM-PCA optimization problem for single cell count data following a negative binomial distribution. (Hint: The negative binomial distribution has two arguments, mean and overdispersion. Only use the matrix factorization UV for one of these two arguments, and set the other as a learned constant.)
- c) What is the PDF of a log-normal distribution? When would you use such a distribution?
- d) Suppose your collaborator generated metabolomics data and wanted you to run GLM-PCA on it. After preprocessing, metabolomics data is often approximated with a log-normal distribution. Write out the GLM-PCA optimization problem for this metabolomics data.
- e) Suppose you asked ChatGPT to answer Question 1d for a brand new, very complicated distribution, and it gave you a specific optimization problem. However, you did not trust ChatGPT's answer and wanted to double-check that it gave you the correct optimization problem. Suggest a way you could do so without re-deriving the optimization problem from scratch.

Problem 2: Canonical correlation analysis (CCA)

We are given two single cell datasets

$$X = [x_{ig}] \in \mathbb{R}^{N \times G} \quad \text{and} \quad Y = [y_{ip}] \in \mathbb{R}^{N \times P}, \quad (1)$$

where each dataset measures a molecular feature (e.g., gene expression, protein expression, chromatin accessibility) on N cells. The two datasets measure different features, with sizes G and P respectively, on the same N cells. Assume X and Y are mean-centered, meaning that $\frac{1}{N} \sum_{i=1}^N x_{ig} = 0$ and $\frac{1}{N} \sum_{i=1}^N y_{ip} = 0$ for all features $g = 1, \dots, G$ and $p = 1, \dots, P$.

Canonical correlation analysis (CCA) is an approach to identify “shared correlation structures” for the two datasets X and Y . Specifically, in CCA, one aims to identify linear combinations

$$u \in \mathbb{R}^G \quad \text{and} \quad v \in \mathbb{R}^P, \quad (2)$$

of the two features such that Xu and Yv are maximally correlated.

a) Fill in the blanks in the CCA optimization problem:

$$\underset{\substack{\|u\|_2=1 \\ \|v\|_2=1}}{\operatorname{argmax}} \operatorname{Corr}(\text{_____,} \text{_____) } \quad (3)$$

b) Suppose $X = Y$ and we replace Corr with Cov in the optimization problem in part a. What problem does this represent? (Hint: We spent a class on it.)

c) Recall the formula for Pearson correlation between two vectors a and b :

$$\operatorname{Corr}(a, b) = \frac{\operatorname{Cov}(a, b)}{\sqrt{\operatorname{Var}(a) \operatorname{Var}(b)}} \quad (4)$$

Rewrite the optimization problem in part **a** using this equation.

d) Define the variables $\Sigma_{XY} = \frac{1}{N-1} X^T Y$, $\Sigma_X = \frac{1}{N-1} X^T X$, and $\Sigma_Y = \frac{1}{N-1} Y^T Y$.

Rewrite the optimization problem in part **c** to remove the variables X and Y , and instead use the variables Σ_{XY} , Σ_X , and Σ_Y .

The variable Σ_{XY} is called the *sample cross-covariance matrix* and the variables Σ_X and Σ_Y are called the *sample covariance matrices* for X and Y , respectively.

Hint: Your optimization problem should have the form

$$\underset{\|v\|_2=1, \|w\|_2=1}{\operatorname{argmax}} \frac{\text{____}}{\sqrt{\text{____}} \sqrt{\text{____}}} \quad (5)$$

e) Define the variables

$$\tilde{u} = \Sigma_X^{1/2} u \quad \text{and} \quad \tilde{v} = \Sigma_Y^{1/2} v. \quad (6)$$

Rewrite the optimization problem in part **d** in terms of the variables $\tilde{u}$, $\tilde{v}$, Σ_X , Σ_Y , Σ_{XY} .

f) Suppose you are told that a solution to the optimization problem

$$\underset{\substack{\|a\|_2=1 \\ \|b\|_2=1}}{\operatorname{argmax}} a^T M b \quad (7)$$

is given by

$$\hat{a} = u \quad \text{and} \quad \hat{b} = v, \quad (8)$$

where u and v are the (matched) leading left and right singular vectors of the matrix M .

Use this knowledge to derive the solution to the optimization problem in part e)

g) *[Open-ended]* The procedure you derived gives you the first *canonical correlation vectors* u_1 and v_1 . How could you extend the approach in part f to find the second canonical correlation vectors u_2 and v_2 ? More generally, how could you find the k -th canonical correlation vectors u_k and v_k for any k ?

h) Consider a different setting where we are given two single cell datasets

$$X \in \mathbb{R}^{N \times G} \quad \text{and} \quad Y \in \mathbb{R}^{M \times G} \quad (9)$$

which measure the same molecular feature (e.g., gene expression) on two cell populations of sizes N and M , respectively.

We would like to use CCA to embed the N cells and M cells, respectively, into a shared embedding space. How could you use CCA to do this?

[CCA is the core algorithmic idea behind the first version of [Seurat](#), a very popular single cell analysis package.]

Problem 3: NMF for multiple samples and modalities

- a) Suppose we measure a single cell data matrix $X \in \mathbb{R}^{N \times P}$ for N cells with P features. We aim to factorize the observed data as $X \approx UV$, where $U \in \mathbb{R}^{N \times K}$ is the factor matrix and $V \in \mathbb{K} \times \mathbb{G}$ is the loading matrix, for some small $K > 0$. Write out the NMF optimization problem for computing this factorization.

- b) Next, suppose we measure S different single cell data matrices $X^{(s)} \in \mathbb{R}^{N_s \times P}$ for $s = 1, \dots, S$. Our data measures the same P features across S different *samples*, i.e. populations of cells. We aim to factorize each sample as $X^{(s)} \approx U^{(s)}V$, where each sample s has its own factor matrix $U^{(s)}$ but all samples share a loading matrix V .

Write out an NMF-style optimization problem for computing this factorization. Qualitatively explain what biological assumption is made if we share the loading matrix V across all samples.

- c) Next, suppose we measure M different single cell data matrices $X^{(m)} \in \mathbb{R}^{N \times P_m}$ for $m = 1, \dots, M$. Our data measures M different molecular modalities (e.g. gene expression, chromatin accessibility, etc.) across the same N cells. We aim to factorize each sample as $X^{(m)} \approx UV^{(m)}$, where each modality m has its own loading matrix $V^{(m)}$ but all modalities share a factor matrix U .

Write out an NMF-style optimization problem for computing this factorization. Qualitatively explain what biological assumption is made if we share the factor matrix U across all samples.

- d) Next, suppose we measure $S \cdot M$ different single cell data matrices $X^{(s,m)} \in \mathbb{R}^{N_s \times P_m}$ for $s = 1, \dots, S$ and $m = 1, \dots, M$. Our data measures M different molecular modalities across S different samples. We aim to factorize each sample as $X^{(s,m)} \approx U^{(s)}V^{(m)}$, where each sample s has its own factor matrix $U^{(s)}$ and each modality m has its own loading matrix $V^{(m)}$.

Write out an NMF-style optimization problem for computing this factorization. Qualitatively explain what biological assumption(s) are being made by this factorization.

- e) In practice, not every modality may be measured in every sample. Let $\Omega \subseteq \{1, \dots, S\} \times \{1, \dots, M\}$ be the set of observed pairs (s, m) of samples and modalities. sample-modality pairs. We only observe single cell data matrices $X^{(s,m)}$ for sample-modality pairs $(s, m) \in \Omega$.

Modify the optimization problem in part **d** to only incorporate the observed data.

- f) Assume that observed pairs are sufficient to learn every factor matrix $U^{(s)}$ and loading matrix $V^{(m)}$. How would you estimate an *unobserved* $X^{(s,m)}$ (i.e. $(s, m) \notin \Omega$)?